

Sri Lanka Institute of Information Technology



Assignment 1

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By

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Summary of the Research Paper: A New Enhanced IGBTQ-based Model for CPU Scheduling

Secure Operating Systems - IE2032

B.Sc. (Hons) in Information Technology Specializing in Cyber Security

Terms of reference

A summary report submitted in fulfilment of the requirement for the module
IE2032 , Sri Lanka Institute of Information Technology.

The report summary is totally based on research paper tile “A New Enhanced
IGBTQ-based Model for CPU Scheduling”

(link for the paper is mentioned at the end of the document)

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1.Introduction to the Topic

This paper addresses a critical aspect of operating systems: CPU scheduling/process scheduling. Efficient CPU scheduling is vital for optimizing processor utilization and ensuring that tasks are completed in a timely manner. As modern computing environments become increasingly complex, the need for effective scheduling algorithms that can handle multiple processes simultaneously has grown. This paper contributes to the field by proposing an enhanced algorithm that builds upon existing models to improve performance metrics such as Average Waiting Time (AWT) and Average Turnaround Time (ATT) and make the operating system more convenient and manage resources and function efficiently in optimal capacity.

2.Problem Addressed

The main problem which is tackled by the authors is the inefficiency of traditional CPU scheduling algorithms, particularly the Round Robin (RR) algorithm. While RR is widely used due to its simplicity and fairness, it often leads to high waiting times and turnaround times, especially in systems with varying process burst times. The authors of this paper highlight that existing enhancements to the RR algorithm have not sufficiently addressed these issues, leading to the need for a more effective solution. This problem is significant because inefficient CPU scheduling can result in increased latency and reduced system performance, impacting user experience and resource utilization, which means hinders the performance of operating system.

3.Proposed Solutions

To address the identified challenges and improve RR efficiency, the authors propose a new algorithm called AE-IGBQT, which stands for the Enhanced Improved Group Based Time Quantum algorithm. This algorithm modifies the traditional RR approach by incorporating the IGBTQ model, which dynamically determines the number of groups into which processes are distributed and adjusts the time quantum accordingly. The primary objective of AE-IGBQT is to reduce both AWT and ATT, thereby enhancing overall CPU scheduling efficiency. The authors assert that their approach outperforms existing algorithms, including the original RR, Dynamic RR with Control Preemption (DRRCP), Group Based Time Quantum (GBTQ), and the previous IGBTQ model.

3.1 Method Used to Address the Problem

Development of the AE-IGBQT Algorithm

- **Enhanced Algorithm Creation:** The authors developed a new algorithm called AE-IGBQT (Enhanced Improved Group Based Time Quantum). This algorithm is designed to modify the existing RR scheduling approach by incorporating the IGBTQ model, which focuses on improving the scheduling efficiency through dynamic adjustments .

Dynamic Grouping of Processes

- **Process Grouping:** A key feature of the AE-IGBQT algorithm is its ability to dynamically group processes. Instead of treating all processes equally, the algorithm categorizes them into groups based on specific criteria. This grouping allows for a more tailored approach to scheduling, where processes with similar characteristics can be managed together, leading to improved performance .

Adaptive Time Quantum Selection

- **Variable Time Quantum:** The AE-IGBQT algorithm introduces an adaptive time quantum mechanism. Unlike the fixed time quantum used in traditional RR scheduling, the AE-IGBQT algorithm adjusts the time quantum based on the characteristics of the processes in each group. This adaptability helps in optimizing the execution time for each process, thereby reducing waiting times and turnaround times .

Experimental Validation

- **Comparative Analysis:** To validate the effectiveness of the AE-IGBQT algorithm, the authors conducted experiments using two different cases with varying numbers of processes. They compared the performance of their proposed algorithm against several existing algorithms, including the original RR, Dynamic RR with Control Preemption (DRRCP), Group Based Time Quantum (GBTQ), and the previous IGBTQ model. The performance metrics measured were Average Waiting Time (AWT) and Average Turnaround Time (ATT) .

Flow chart of AE-IGBTQ algorithm

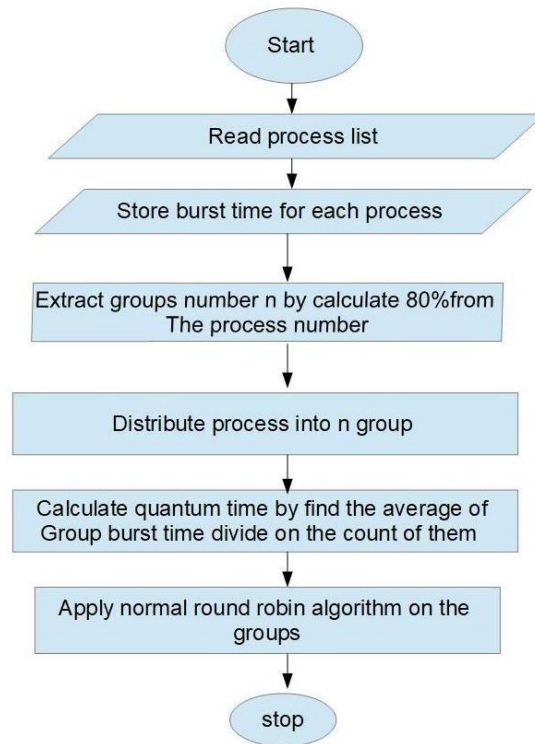


Figure 2. Flowchart of authors' proposed AE-IGBQT method.

4. Methodology Used for Experimentation

The authors conducted experiments using two distinct cases with varying numbers of processes to evaluate the performance of the AE-IGBQT algorithm. They compared the results against those obtained from the beforementioned algorithms. The experimental setup involved determining burst times for the processes based on previous studies, allowing for a controlled comparison. The performance metrics of interest were AWT and ATT, which were measured to assess the effectiveness of the proposed algorithm. The results were presented in tables and figures to illustrate the improvements achieved by AE-IGBQT.

5. Results and Conclusions

The results of the experiments demonstrated that the AE-IGBTQ algorithm significantly reduced both AWT and ATT compared to the other algorithms tested. For instance, in one case, the algorithm achieved an AWT of 172.625 and an ATT of 237.125, showcasing its effectiveness in improving CPU scheduling performance.

- Results obtained from 8 process test case

PID	BT	Algorithms	QT	AWT	ATT
1	61	RR	20	430.5	495.0
2	62	DRRCP	61, 62, 63, 64, 65, 66, 67, 68	220.5	285.0
3	63	GBTQ	20,20,20,20	430.5	495.0
4	64	(Panda et al. 2014)			
5	65	IGBTQ	61, 62, 63, 64, 65, 66, 67, 68	220.5	285.0
6	66	(Mijinyawa & Abdullahi 2017)			
7	67	AE-IGBTQ	61, 62, 63, 64, 65, 66, 67, 68	172.63	237.13
8	68	(proposed algorithm)			

Figure 2. Data of Case 1 & Summary results of Case 1.

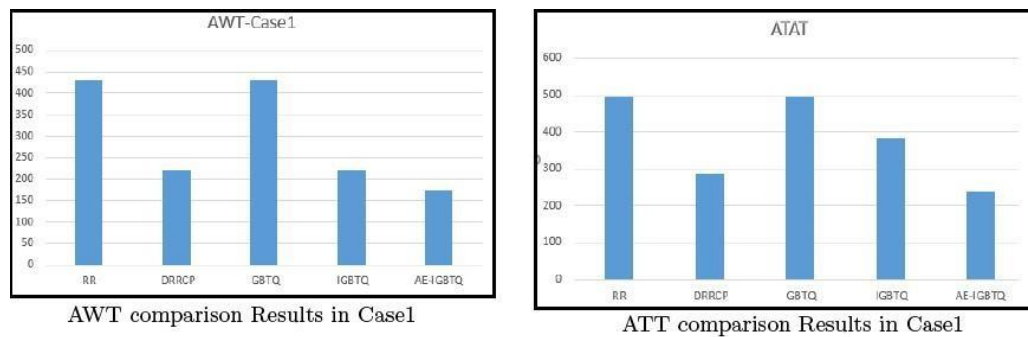


Figure 3. AWT & ATT results from Case 1.

- Test results obtained from 7 process test case

$P_i D$	BT	Algorithms	QT	AWT	ATT
1	110	RR	20	573.8	682.5
2	89	DRRCP	08, 86, 108, 95	367.7	476.4
3	113	GBQT	20, 137, 20.	524.1	632.9
4	137	IGBQT	110, 89, 113, 137, 86, 131, 95.	324.4	433.1
5	86	AE-IGBQT (proposed algorithm)	110, 89, 113, 137, 86, 131, 95.	255.71	364.43
6	131				
7	95				

Figure 4. Data of Case 2 & Summary results of Case 2

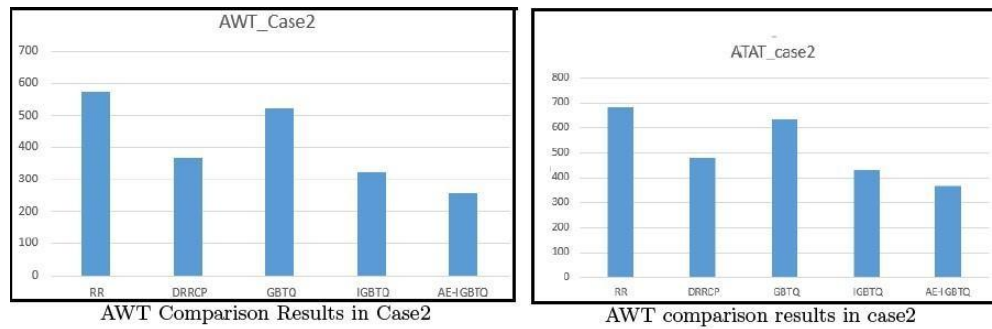


Figure 4. AWT & ATT results from Case 2.

In conclusion this algorithm has achieved ,

1. Development of AE-IGBQT

- **Successful Enhancement:** The authors successfully developed the AE-IGBQT algorithm, which modifies the traditional Round Robin (RR) scheduling algorithm by incorporating the IGBTQ model. This enhancement aims to address the inefficiencies associated with CPU scheduling .

2. Improvement in Performance Metrics

- **Reduction in AWT and ATT:** The primary objective of the AE-IGBQT algorithm was to improve the Average Waiting Time (AWT) and Average Turnaround Time (ATT). The results demonstrated that the AE-IGBQT algorithm significantly reduced these metrics compared to previous algorithms, including the original RR, Dynamic RR with Control Preemption (DRRCP), Group Based Time Quantum (GBTQ), and the IGBTQ algorithm itself. This indicates a marked improvement in scheduling efficiency .

3. Dynamic Process Management

- **Dynamic Grouping and Quantum Time:** The AE-IGBQT algorithm's success is attributed to its dynamic determination of the number of groups in which processes are distributed, as well as the adaptive selection of quantum time (QT). This flexibility allows for better management of processes, leading to enhanced performance outcomes .

4. Future Work Directions

- **Potential for Further Improvements:** The authors express their intention to explore further enhancements to other CPU scheduling algorithms in future work. This suggests that the AE-IGBQT algorithm could serve as a foundation for developing even more efficient scheduling methods in the future .

6. Critical Evaluation of the Paper

The paper presents several strengths, including:

- **Innovative Approach:** The AE-IGBQT algorithm represents a novel enhancement to existing scheduling models, addressing a well-known problem in CPU scheduling.
- **Comprehensive Testing:** The authors conducted thorough experiments, comparing their results against multiple established algorithms, which adds credibility to their findings.
- **Clear Presentation:** The results are well-organized, with tables and figures that effectively communicate the improvements achieved.

However, there are also some weaknesses:

- **Limited Scope:** The experiments were conducted with a specific set of processes, which may not fully represent the diversity of workloads encountered in real-world scenarios.
- **Lack of Real-World Application:** The paper does not discuss how the proposed algorithm could be implemented in actual operating systems, which would enhance its practical relevance.

7. Broader Concepts in Operating Systems

The concepts explored in this paper relate closely to the broader principles of operating systems, particularly in the context of process management and resource allocation. The findings contribute to a deeper understanding of how scheduling algorithms can be optimized to improve system performance.

- **Relevance to Modern Operating Systems:** The principles discussed in the paper are highly relevant to modern operating systems, which must efficiently manage multiple processes and resources. As systems become more complex, the need for advanced scheduling algorithms that can adapt to varying workloads is crucial.
- **Application and Extension:** The ideas presented in the paper could be applied to current and future operating systems by integrating the AE-IGBQT algorithm into existing scheduling frameworks. Additionally, further research could explore the algorithm's adaptability to different types of workloads, including real-time systems and cloud computing environments.

8. References

This summary report is solely based on the research paper titled “A New Enhanced IGBTQ-Based Model for CPU Scheduling.”

The link for the paper : - Ra'ed, M., Al-Khatib., Asef, Al-Khateeb., Esraa, Al-Daom., Ishraq, T., Al-Dagamseh., Ayat, A. Tawalbeh., Laith, Abualigah. (2023). A New Enhanced IGBTQ-based Model for CPU Scheduling. Applied and Computational Engineering, 8(1), 411-417.

Available from: doi:10.54254/2755-2721/8/20230207 Or

Link from research gate :-

https://www.researchgate.net/publication/372822466_A_New_Enhanced_IGBTQ-based_Model_for_CPU_Scheduling